
Query-Only Attention Preserves Plasticity in Fully Online Continual Learning

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Abstract

Continual learning in fully online settings is challenging due to non-stationary data streams and single-pass training without task boundaries and no complete data reuse. In this regime, models often suffer from a *loss of plasticity*, i.e., a reduced ability to adapt to new data over time, leading to degraded forward performance. We propose Query-Only Attention (QOA), a simplified attention-based approach that learns a direct similarity function between queries and past examples, enabling consistent query–support matching without relying on keys or values. We evaluate QOA on long-horizon online variants of CIFAR-100, ImageNet, and additional benchmarks. Existing continual learning methods, including replay, regularization, and re-initialization approaches, often exhibit degraded forward performance under strict online, non-stationary conditions due to slow adaptation. In contrast, QOA achieves higher prequential accuracy and near-zero cumulative regret, indicating faster and more reliable adaptation. We provide theoretical insight suggesting that QOA encourages convergence toward a shared solution across tasks, stabilizing adaptation and mitigating loss of plasticity.

1 Introduction

Continual learning remains a fundamental challenge in deep learning [36], where models must learn from non-stationary data streams without succumbing to catastrophic forgetting [18] or loss of plasticity [11]. In fully online settings, where data is observed only once and task boundaries are absent, models must continuously adapt to evolving distributions while maintaining strong forward performance.

While existing approaches mitigate forgetting using replay buffers [39] or regularization [18], preserving plasticity, the ability to acquire new knowledge over time remains less understood. Moreover, many methods are primarily evaluated in settings that allow multiple passes over data or rely on task boundaries, and emphasize final performance metrics. As a result, they often overlook the importance of rapid adaptation, where slow convergence can incur significant cumulative error in fully online, non-stationary settings.

Recent work suggests that attention mechanisms [33, 28, 16] can support stable adaptation across tasks [35]. However, why attention is able to mitigate loss of plasticity in non-stationary settings remains poorly understood. This raises a key question: which components of attention are essential for enabling continual adaptation, and can this behavior be recovered with a simpler mechanism?

To investigate this, we propose Query-Only Attention (QOA), a simplified attention-based model that removes keys and values and instead learns a direct similarity function between queries and support examples. By isolating the query-support interaction, we aim to identify the minimal structure required for stable adaptation. We provide a theoretical analysis showing that this interaction induces behavior that closely approximates full attention under suitable conditions, suggesting that the key-

value structure is not necessary to recover the core functional behavior of attention in continual learning.

Empirically, QOA achieves state-of-the-art performance across a range of challenging continual learning benchmarks, outperforming strong baselines including replay-based and regularization-based methods. In addition, QOA exhibits faster adaptation dynamics, leading to improved performance in fully online settings where slow convergence results in accumulated errors. Our analysis further shows that QOA maintains stable task-adapted representations and exhibits reduced parameter drift over long horizons, enabling sustained plasticity.

Key contributions:

- We derive *Query-Only Attention* (QOA) from the standard attention mechanism [33] for the online continual learning setting. By removing the key and value matrices and using direct query–support similarity with support-label aggregation, QOA simplifies the architecture while retaining the essential interaction structure of attention, leading to faster convergence and improved performance in fully online, non-stationary settings.
- We provide a theoretical analysis explaining why both full attention and query-based models can mitigate loss of plasticity. We show that their objectives induce consistent query–support matching across tasks, leading to convergence toward a shared *global* solution rather than task-specific local solutions.
- We show that QOA is computationally better suited to online continual learning than full attention. In particular, QOA reduces the dependence on buffer size from quadratic to linear by avoiding pairwise interactions within the buffer, making it more scalable in fully online settings.
- We conduct a comprehensive empirical comparison in difficult long-horizon, non-stationary online settings against replay-, regularization-, activation-, and re-initialization-based continual learning baselines. QOA achieves superior prequential accuracy, showing stronger online performance and sustained plasticity.
- We evaluate convergence speed using relative cumulative regret. QOA achieves near-zero regret on several classification benchmarks, indicating faster convergence and more reliable adaptation compared to existing continual learning methods.
- We introduce a task-wise augmentation protocol for large-scale benchmarks (CIFAR-100, Tiny ImageNet, and ImageNet-1000), where each task is generated via deterministic, per-task augmentations. This induces controlled distribution shifts across tasks, enabling realistic evaluation of online adaptation and plasticity without relying on repeated data or explicit task boundaries.
- As supporting analysis, we include Hessian rank evaluation [24] and comparisons with gradient-based meta-learning methods [13], which provide additional evidence for the role of global solution convergence in preserving plasticity.

2 Related Work

Deep neural networks typically assume stationary data and multiple passes over the dataset, whereas online continual learning operates on non-stationary streams where each sample is observed once. This setting introduces two fundamental challenges: *catastrophic forgetting*, the degradation of previously learned tasks [27, 17, 29], and *loss of plasticity*, the reduced ability to learn new tasks over time [11, 1, 8], which is commonly observed in standard training methods such as *vanilla backpropagation* (BP).

Several recent methods explicitly target loss of plasticity. *Continual Backpropagation* (CBP) [11] restores plasticity through selective re-initialization, *Concat ReLU* (CReLU) [1] improves representational capacity through activation design, and *Regenerative Regularization* (RegenReg) [22] encourages parameters to remain adaptable. Regularization methods such as *Elastic Weight Consolidation* (EWC) [18], *Synaptic Intelligence* (SI) [38], and *Learning without Forgetting* (LwF) [25], as well as replay methods such as *Experience Replay* (ER) [30] and *Dark Experience Replay++* (DER++) [4], are also widely used to stabilize continual learning.

However, many of these methods are evaluated in relatively controlled settings, often with clear task boundaries or evaluation focused on end-of-task accuracy. End-of-task evaluation can overstate adaptability because it measures performance after the model has already trained on the current task. In strict online settings with distribution shifts across tasks, the more relevant question is whether the model adapts quickly as data arrives. We therefore emphasize prequential accuracy and cumulative regret, which directly measure online adaptability and convergence speed.

Recent work suggests that attention-based models can mitigate loss of plasticity in controlled or toy settings [35]. However, the mechanisms underlying this behavior are not well understood. In this work, we provide a simple theoretical perspective, supported by empirical analysis, explaining why similarity-based interactions in attention help preserve plasticity.

Building on this insight, we derive Query-Only Attention (QOA), a simplified formulation tailored for online continual learning. QOA retains the query-support matching mechanism of attention while removing keys and learned value vectors, reducing computational cost from $\mathcal{O}(n^2d)$ to $\mathcal{O}(nd)$. We provide evidence that both full attention and query-based models encourage convergence toward a shared representation across tasks, leading to more stable learning and improved plasticity. Empirically, QOA achieves improved cumulative regret and prequential accuracy in more realistic, non-stationary settings, demonstrating stronger online performance under distribution shift.

More broadly, our approach connects attention mechanisms, in-context learning [12, 37, 2], and meta-learning frameworks such as *Model Agnostic Meta-Learning (MAML)* [13] and metric-based methods [34, 19, 32]. Finally, to study plasticity dynamics, we adopt curvature-based analyses such as effective Hessian rank [24, 31], where stable rank indicates sustained adaptability.

3 Problem Statement

We consider an *online continual learning* setting where a model observes stream of data sequentially, each sample seen once (no epochs or replay) [5, 3]. The stream is generated from tasks $\{T_1, T_2, \dots\}$, where each task T_i follows a (potentially non-stationary) distribution $\mathcal{D}_i(x, y)$. The model is trained online without access to task identities or boundaries and updating parameters using only current observations (single samples or small batches) [20, 7, 3]. Evaluation is performed in a forward (prequential) manner along the stream, *where each sample is first used for testing before training*, reflecting real-world deployment scenarios [9, 14, 5, 26]. We measure **forward performance (plasticity)** via predictive accuracy on incoming samples prior to training, where degradation over time indicates loss of plasticity.

4 Method

4.1 From Attention to Query-Only Attention

We derive Query-Only Attention (QOA) from standard attention in the online continual learning setting. Let τ denote the current task and $x_{\tau,t} \in \mathbb{R}^{1 \times d}$ the current datapoint. Let the support set contain m examples $\mathcal{S}_\tau = \{(x_{\tau,j}, y_{\tau,j})\}_{j=1}^m$ with $x_{\tau,j} \in \mathbb{R}^{1 \times d}$. In the fully online setting, data arrives as a stream, and the support set $\mathcal{S}_\tau$ corresponds to a buffer of fixed size m that stores the most recent m previously seen examples in a first-in-first-out (FIFO) manner. We refer to these previously observed elements in the buffer as the support set.

Full attention stacks the datapoint and support set into

$$X_{\tau,t} = \begin{bmatrix} x_{\tau,t} \\ x_{\tau,1} \\ \vdots \\ x_{\tau,m} \end{bmatrix} \in \mathbb{R}^{(m+1) \times d},$$

and computes $Q = X_{\tau,t}W_Q$, $K = X_{\tau,t}W_K$, and $V = X_{\tau,t}W_V$, where $W_Q, W_K, W_V \in \mathbb{R}^{d \times d}$.

The attention matrix is $A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) \in \mathbb{R}^{(m+1) \times (m+1)}$, and the output is $Y = AV \in \mathbb{R}^{(m+1) \times d}$.

This computes representations for both the current data-point and all support elements. However, in our setting we only require the prediction for $x_{\tau,t}$, making support–support interactions unnecessary.

Let $n = m + 1$. The projection step costs $\mathcal{O}(nd^2)$, while computing $QK^\top$ and AV each costs $\mathcal{O}(n^2d)$. Thus, full attention scales as $\mathcal{O}(nd^2 + n^2d)$, with quadratic dependence on the support size through the pairwise attention terms. For predicting the current data-point, the only required interactions are between $x_{\tau,t}$ and each support example $x_{\tau,j}$. In standard attention, this interaction is given by

$$q_{\tau,t}k_{\tau,j}^\top = (x_{\tau,t}W_Q)(x_{\tau,j}W_K)^\top.$$

QOA retains the query–support interaction structure but replaces the projected dot-product similarity with a learned similarity function.

$$f_\theta(x_{\tau,t}, x_{\tau,j}) \in \mathbb{R}.$$

Computing this score for all support examples gives the task-adapted representation

$$\theta'_{\tau,t} = [f_\theta(x_{\tau,t}, x_{\tau,1}), \dots, f_\theta(x_{\tau,t}, x_{\tau,m})] \in \mathbb{R}^m.$$

Unlike standard attention, QOA does not apply softmax normalization to these scores.

Prediction via label aggregation. In standard attention, the output is a weighted sum of learned value vectors. In online continual learning, each support example already has an associated label or target. Therefore, QOA replaces learned value vectors with the corresponding support labels. Each support example contributes its label to the prediction according to its learned similarity with the current datapoint, yielding a similarity-weighted label aggregation.

For classification, let

$$Y_\tau^{\text{sup}} = [e(y_{\tau,1}); \dots; e(y_{\tau,m})] \in \mathbb{R}^{m \times C},$$

where $e(y_{\tau,j})$ is the one-hot encoding of label $y_{\tau,j}$ and C is the number of classes. The predicted logits are

$$Y_{\text{pred}} = \theta'_{\tau,t} Y_\tau^{\text{sup}} \in \mathbb{R}^{1 \times C}.$$

For regression, let

$$y_\tau^{\text{sup}} = [y_{\tau,1}, \dots, y_{\tau,m}]^\top \in \mathbb{R}^m.$$

The predicted output is

$$Y_{\text{pred}} = \theta'_{\tau,t} y_\tau^{\text{sup}} \in \mathbb{R}^1.$$

Thus, QOA preserves the attention principle of aggregating support information based on query–support similarity, while removing support–support interactions, explicit key projections, and learned value vectors. Focusing on scaling with support size, QOA computes m query–support scores and scales linearly as $\mathcal{O}(md)$, whereas full attention scales as $\mathcal{O}(nd^2 + n^2d)$ with $n = m + 1$, exhibiting quadratic dependence on the support size. *This difference is significant in online settings, as it allows QOA to operate with substantially larger support sets while maintaining lower computational cost, enabling more effective use of past data.*

Balanced support sampling. In practice, we do not use all elements in the buffer to compute Y_{pred} . Instead, we construct a support set by sampling a fixed number of examples per class. For a classification task with C classes, we sample k examples per class, even if the buffer is significantly larger. This balanced sampling ensures that each class contributes equally to the prediction and avoids bias toward overrepresented classes. It also aligns with the query–support matching structure of QOA, where each class provides comparable evidence during similarity-based aggregation, while reducing computational cost.

Training Procedure. The model is trained in a fully online manner. At each step, the current sample is added to the buffer using a FIFO policy, a support set is sampled, and the model is updated via gradient descent on the prediction loss.

Algorithm 1 Query-Only Attention (QOA) with Replay Buffer

Input: Data stream $(x_{\tau,t}, y_{\tau,t})$; buffer $\mathcal{B}$ of size n ; support size m ; scoring network f_θ

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1: Initialize parameters  $\theta$ ; buffer  $\mathcal{B} \leftarrow \emptyset$ 
2: for each timestep  $t$  with sample  $(x_{\tau,t}, y_{\tau,t})$  do
3:    $\mathcal{S}_\tau = \{(x_{\tau,j}, y_{\tau,j})\}_{j=1}^m \subset \mathcal{B}$ 
4:   Query-support scoring:
5:   Stack supports:  $X_\tau^{\text{sup}}$ 
6:   Repeat query:  $X_{\tau,t}^{\text{rep}}$ 
7:   Concatenate:  $Z = [X_{\tau,t}^{\text{rep}}, X_\tau^{\text{sup}}]$ 
8:    $\theta'_{\tau,t} = f_\theta(Z)$ 
9:   Aggregate prediction:
10:  if classification then
11:     $\hat{y}_{\tau,t} = (\theta'_{\tau,t})^\top Y_\tau^{\text{sup}}$ 
12:  else
13:     $\hat{y}_{\tau,t} = (\theta'_{\tau,t})^\top y_\tau^{\text{sup}}$ 
14:  end if
15:  Loss and update:
16:   $\mathcal{L}_{\tau,t} \leftarrow \ell(\hat{y}_{\tau,t}, y_{\tau,t})$ 
17:   $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\tau,t}$ 
18:  Update buffer:
19:  Insert  $(x_{\tau,t}, y_{\tau,t})$  into  $\mathcal{B}$ 
20:  Evict oldest if  $|\mathcal{B}| > n$ 
21: end for
```

Note: The current sample is not included in the support set. The buffer is updated after learning to avoid trivial matching.

5 Experiments

5.1 Experimental Setup

We evaluate our method on five benchmarks: Slowly Changing Regression (SCR), Permuted MNIST, CIFAR-100, Tiny ImageNet, and ImageNet, under a strict online continual learning setting.

CIFAR-100, Tiny ImageNet, and ImageNet. In CIFAR-100 [21] and Tiny ImageNet [23], each task consists of 5 classes sampled uniformly at random (with replacement) from the global label set. To introduce distributional shift, we apply a fixed augmentation per task (e.g., cropping, rotation, or color transformations) to all images. This ensures that even when the same underlying image is revisited, the model does not observe identical inputs. This design induces two key challenges: (i) *recurring classes*, where the same class reappears across tasks, and (ii) *distributional variation*, where the input distribution for a class changes across tasks. As a result, the model must continuously adapt to new variations of previously seen classes without explicit task information. For ImageNet [10], we adopt a similar setup with binary classification tasks. Each task consists of 2 classes sampled from the 1000-class dataset with task-specific augmentations. This scales the problem to a larger and more diverse dataset while retaining the challenges of recurring classes and distribution shift. We train CIFAR-100 and Tiny ImageNet for 10,000 tasks and ImageNet for 5,000 tasks due to computational constraints.

Permuted MNIST. We use the benchmark from [11] where each task consists of all 10 classes, with a task-specific random permutation applied to the input pixels. This yields tasks that share the same label space but differ in input distribution, requiring the model to adapt to changing representations while maintaining classification performance [15]. We train for 10,000 tasks.

Slowly Changing Regression (SCR). We use the SCR benchmark from [11], where the data distribution evolves gradually over time. This setting isolates the effect of non-stationarity and is specifically designed to study loss of plasticity. We train for 1,000 tasks.

Evaluation Protocol and Metrics. We consider a fully online continual learning setting where data is presented as a continuous stream without task identifiers, and each sample is observed only once. Tasks are constructed to exhibit both *distributional shift* and *recurring classes*, creating a challenging non-stationary environment. Unlike standard benchmarks that rely on task boundaries or multiple epochs over data, our setup enforces single-pass learning with evolving input distributions. We evaluate models using *prequential accuracy*, where each sample is first used for testing before the model is updated [9]:

$$\text{Acc}_T^{(m)} = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(\hat{y}_t^{(m)} = y_t),$$

where $\hat{y}_t^{(m)}$ is the prediction of method m at time t , and T denotes the length of a task. For each task, prequential accuracy is computed as the average of prequential evaluations over all time-steps belonging to that task. Under this protocol, performance on a new task is initially low and improves as more samples are observed. Consequently, prequential accuracy is typically lower than evaluation based on held-out test C sets measured after training on a task, as it more closely reflects real-world deployment where predictions must be made before adaptation. Importantly, this metric explicitly penalizes slow adaptation, since models incur errors over the early portion of each task. We also report held-out accuracy, measured at the end of each task on a separate test set. These results, along with additional plots not included in the main paper due to space constraints, are provided in Appendix C.

Baselines. We first consider methods explicitly designed to address loss of plasticity, including *Continual Backpropagation (CBP)*, *Concat ReLU (CReLU)*, and *Regenerative Regularization (RR)*. While these approaches aim to preserve adaptability, they have not been evaluated under prequential metrics, which directly penalize slow adaptation. We additionally compare against a *full attention* model, from which our formulation is derived, to isolate the impact of the query-only design. Finally, we include replay-based methods such as *Experience Replay (ER)* and *Dark Experience Replay++ (DER++)*, which are strong baselines for mitigating catastrophic forgetting, to evaluate how they perform under our strictly online, non-stationary setting.

5.2 Experiment Results

Table 1: Prequential accuracy across datasets, with *start* and *end* denoting early-stage and long-horizon performance averaged over the first and last 100 tasks, respectively; accuracy ($\uparrow$) is higher-is-better, while SCR loss ($\downarrow$) is lower-is-better.

Method	PMNIST		CIFAR-100		TinyIN		ImageNet		SCR	
	Start $\uparrow$	End $\uparrow$	Start $\uparrow$	End $\uparrow$	Start $\uparrow$	End $\uparrow$	Start $\uparrow$	End $\uparrow$	Start $\downarrow$	End $\downarrow$
Vanilla BP	65.42	53.44	27.69	21.09	22.17	20.52	52.73	52.41	0.970	0.641
CBP	62.58	67.09	38.84	51.57	29.87	35.10	61.68	55.74	1.094	0.693
Concat ReLU	74.72	78.98	28.54	44.15	23.53	25.21	54.09	55.10	0.814	0.647
Dark Exp++	73.88	64.15	28.28	33.59	23.27	25.66	53.64	59.25	0.961	0.447
ER	71.38	64.44	30.49	24.00	23.71	21.37	52.78	59.19	0.995	0.512
EWC	49.84	39.09	27.86	65.17	22.45	45.47	53.07	59.44	2.761	2.897
Full Attention	29.70	88.66	37.51	84.35	29.22	62.25	59.57	76.16	1.118	0.794
QOA (Ours)	80.86	87.81	47.55	88.88	36.59	67.78	65.98	85.97	7.542	0.510
Reg. Reg.	66.80	68.13	27.05	21.93	22.38	19.43	52.70	52.44	1.006	0.701

Observations on the performance Across classification benchmarks, **Query-Only Attention (QOA)** consistently outperforms all baselines, with full attention as the closest competitor; the only exception is Permuted MNIST, where full attention performs slightly better (by $<1\%$). QOA achieves higher final performance, faster convergence, and better scalability on harder tasks. In regression, QOA performs slightly worse than DER, likely due to high initial error. A key observation is that methods explicitly designed to mitigate *loss of plasticity*, such as CBP, CReLU, and RR, often maintain adaptability but exhibit slow adaptation. Under the prequential accuracy evaluation protocol this results in poor performance, as these methods incur errors over a large number of early time-steps within each task before improving. For example, in CIFAR-100 and Tiny ImageNet, each task spans many time-steps, and slow initial adaptation leads to significantly lower average accuracy. Replay-based methods such as ER and DER++, while effective at mitigating forgetting, struggle in this

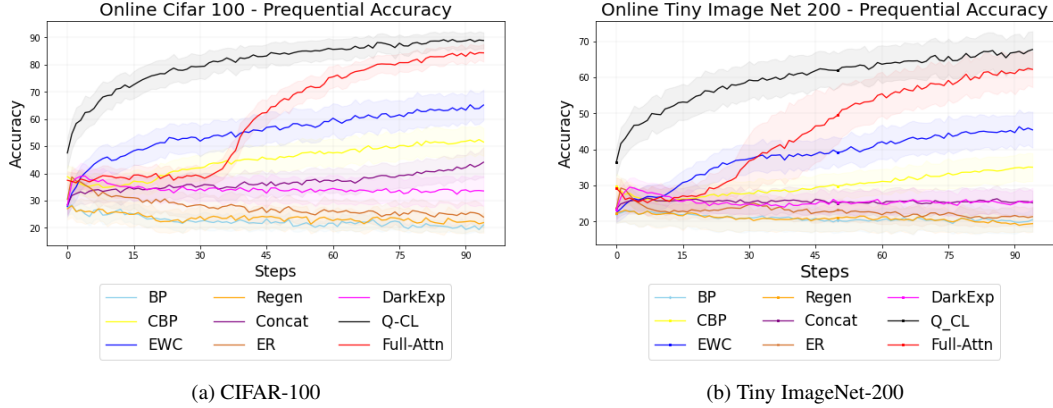


Figure 1: Prequential accuracy over 10^4 tasks. Each point is averaged over 100 tasks. QOA achieves consistently higher accuracy and faster adaptation compared to baselines. We present prequential accuracy plots for CIFAR-100 and Tiny ImageNet in the main text. Corresponding results for all other datasets, along with forward performance plots, are deferred to the C due to space limitations.

setting due to both loss of plasticity and limited ability to rapidly adapt to evolving distributions. As a result, their performance remains suboptimal under strict online conditions. In contrast, QOA and full-attention models achieve strong performance by rapidly adapting to new tasks while maintaining stability over long horizons. We attribute this to the emergence of a *global solution*, where task-level representations align across tasks. Once this shared structure is learned, performance remains stable even at the beginning of new tasks, enabling both fast adaptation and sustained accuracy. These results highlight a fundamental limitation of existing approaches: preserving plasticity alone is insufficient in fully online settings, as methods must also achieve rapid adaptation; prequential evaluation exposes this trade-off, and QOA is able to satisfy both simultaneously.

Convergence Speed via Regret. While prequential performance captures forward adaptation, it does not fully reflect the time required to reach a stable solution. As observed in our experiments, once a *global solution* is reached, models perform consistently well on subsequent tasks. However, reaching this regime can require many initial tasks, during which performance remains suboptimal and accumulates error. To capture this effect, we measure convergence speed using *relative cumulative regret* following standard online learning formulations [6]. Let $\ell_t^{(m)}$ denote the loss of method m at time t , and define the best instantaneous loss as $\ell_t^* = \min_{m'} \ell_t^{(m')}$. The regret of method m over T steps is

$$R_T^{(m)} = \sum_{t=1}^T (\ell_t^{(m)} - \ell_t^*),$$

where lower regret indicates faster convergence and improved online adaptation. We use task-appropriate loss functions (e.g., cross-entropy for classification and mean squared error for regression).

Table 2: Relative cumulative regret (lower is better); QOA achieves near-zero regret, indicating rapid convergence to a shared solution, while baselines converge more slowly.

Method	PMNIST ↓	CIFAR-100 ↓	TinyIN ↓	ImageNet ↓	SCR ↓
Vanilla BP	2859.33	5547.38	3703.83	1445.60	10.08
CBP	2027.32	3427.54	2841.49	1247.24	12.10
Concat ReLU	839.95	4177.70	3265.27	1348.52	8.97
Dark Exp.	1905.97	4422.49	3268.27	1171.60	1.75
ER	1710.34	5042.10	3521.32	1065.16	4.70
EWC	4393.06	2430.45	2070.90	1250.68	118.07
Full Attention	435.69	1933.06	1409.17	1072.36	22.09
QOA (Ours)	22.34	0.00	0.00	0.00	10.89
Reg. Reg.	1877.02	5432.98	3705.72	1443.58	13.50

Regret analysis. We analyze relative cumulative regret across datasets (lower is better). QOA consistently achieves the lowest regret on classification benchmarks, often approaching zero, indicating rapid convergence to a stable solution. In contrast, baselines exhibit substantially higher regret,

reflecting slow adaptation under non-stationary conditions. Notably, while the full-attention model achieves competitive final performance 5.2, it incurs significantly higher regret compared to QOA. This indicates that full attention requires substantially more time to converge, resulting in prolonged periods of suboptimal performance during early time-steps. In fully online settings, where predictions must be made before adaptation and errors accumulate over time, such delayed convergence can have a significant impact on overall performance. These results highlight that convergence speed is a critical factor in online continual learning: methods that eventually achieve strong performance may still be ineffective in practice if they adapt too slowly. On SCR, DER++ achieves the lowest regret, while QOA remains a close second. As noted above, QOA exhibits higher initial error in this setting, which disproportionately increases cumulative regret despite strong performance after adaptation. This suggests that, in regression settings with gradually evolving distributions, early-stage behavior plays a more significant role in regret, highlighting distinct optimization dynamics compared to classification benchmarks.

6 Task-Level Representations and Global Solutions

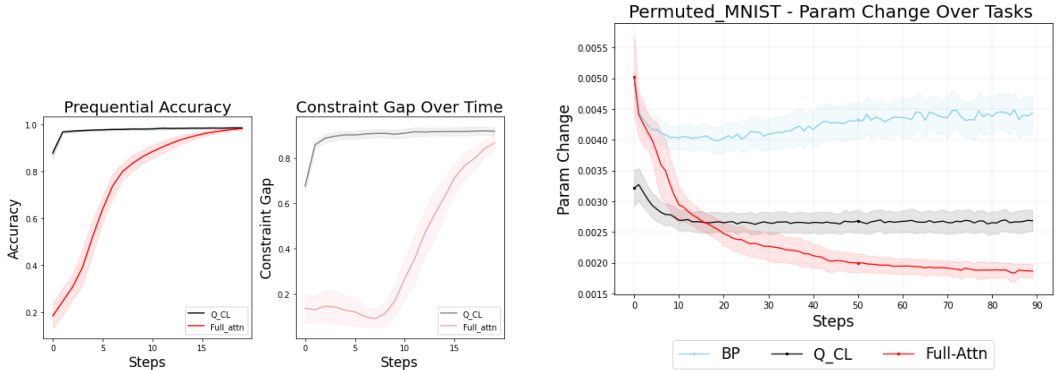
In QOA, predictions are formed via similarity-weighted aggregation over a support set, yielding a task-level representation $\theta'_{\tau,t}$.

To correctly predict $y_{\tau,t}$, the model must assign larger values in $\theta'_{\tau,t}$ to supports with matching labels than to those with non-matching labels. Formally,

$$\mathbb{E}_{j:y_j=y_{\tau,t}} \theta'_{\tau,t,j} > \mathbb{E}_{j:y_j \neq y_{\tau,t}} \theta'_{\tau,t,j}.$$

This induces a task-invariant relational structure, as the constraint depends only on label agreement and not on task identity. As a result, the representations $\theta'_{\tau,t}$ tend to align across tasks, leading to a shared global solution that stabilizes learning and mitigates loss of plasticity.

Full derivation is provided in Appendix B.



(a) Consistency of task-adapted representations (θ') over 2K Permuted MNIST tasks

(b) Parameter change between successive tasks over 10K Permuted MNIST tasks

Figure 2: Model dynamics across tasks. (a) QOA and full-attention increasingly satisfy the matching constraint, leading to consistent task-adapted representations. (b) Parameter updates decrease over time, indicating convergence to a stable solution.

Empirical verification of global solution. We empirically validate the global solution hypothesis using two complementary analyses on Permuted MNIST 2a and 2b. First, we track the *constraint gap*, defined as the difference between similarity assigned to supports with matching and non-matching labels. As shown in figure 2a both QOA and full-attention exhibit a steadily increasing gap over time, indicating progressive satisfaction of the task-invariant matching constraint. This closely correlates with improvements in prequential accuracy, linking constraint satisfaction to faster adaptation. Second, we measure parameter change across tasks by computing the distance between parameters at the end of successive tasks. As shown in Figure 2b, QOA and full-attention exhibit decreasing parameter updates, indicating convergence to a stable solution. In contrast, vanilla backpropagation maintains large parameter shifts, reflecting continual re-adaptation and lack of convergence. Together, these results support the global solution hypothesis: enforcing the matching constraint leads to aligned task-level representations, stable learning dynamics, and reduced loss of plasticity.

Comparison to gradient-based adaptation. Gradient-based meta-learning methods such as MAML compute task-specific parameters

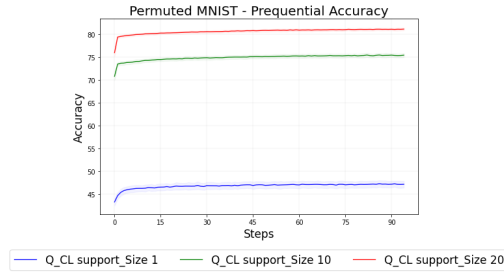
$$\theta'_t = \theta - \alpha \nabla_{\theta} \mathcal{L}_t(\theta), \quad (1)$$

where $\mathcal{L}_t$ depends on the current task.

Unlike query-based or attention models, MAML does not induce a task-invariant structural constraint on the relationship between inputs. The adaptation step is driven entirely by task-specific gradients, which vary across tasks. As a result, the resulting representations θ'_t are not required to align or converge across tasks. In contrast, our formulation induces a consistent matching structure across tasks through the learning objective, enabling convergence to a shared task-level representation. The empirical results are shown in Fig. 4.

6.1 Ablation Study: Effect of Buffer Size

Here, we study the impact of buffer size on performance by varying the number of support samples n while keeping all other factors fixed. As shown in Figure 3, performance improves consistently as the buffer size increases, indicating that the model effectively leverages additional past information. The gains saturate beyond a certain point, suggesting diminishing returns for very large buffers. This behavior is consistent with the linear scaling of our method with respect to buffer size, enabling efficient utilization of larger buffers.



(a) Ablation study on Permuted MNIST

Figure 3: Prequential accuracy over 10^4 tasks. Each point is averaged over 100 tasks. The accuracy improves as the support size increases.

7 Conclusion

We proposed Query-Only Attention (QOA) for fully online continual learning, targeting the loss of plasticity under non-stationary data streams without task boundaries or repeated data access. QOA replaces standard attention with a learned similarity (distance) function between a query and support examples. We showed both empirically and via theoretical analysis that this constraint leads to the emergence of a shared *global solution*, where task-level representations align across tasks. This behavior is also observed in full attention; however, QOA achieves this alignment more efficiently due to its simplified structure. As a result, QOA converges faster and more reliably to the global solution, as reflected in consistently lower cumulative regret. Across challenging online benchmarks, QOA achieves higher prequential accuracy and lower regret than a range of continual learning baselines, demonstrating faster adaptation and improved forward performance. Our analysis further highlights that convergence speed is critical in online settings. Methods that eventually perform well can still incur substantial errors if they adapt slowly. In contrast, QOA rapidly satisfies the matching constraint and stabilizes representations early, reducing the need for continual parameter updates. While QOA relies on a support set, its simplicity and linear scaling with support size make it practical for long-horizon learning. Future work includes extending the approach to larger-scale settings, reducing dependence on explicit support sets, and developing a deeper theoretical understanding of the global solution regime.

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A Technical appendices and supplementary material

A.1 Broader impacts

This work focuses on foundational aspects of online continual learning, specifically understanding and mitigating loss of plasticity in non-stationary settings. By improving adaptation speed and preventing the need for repeated training, our approach can contribute to more efficient and scalable learning systems. Potential positive impacts include reduced computational cost, improved energy efficiency, and more adaptive models in applications such as robotics, healthcare monitoring, and personalized systems.

However, continual learning systems that adapt over time may also introduce risks. In particular, adaptive models deployed in sensitive domains (e.g., surveillance or user profiling) may amplify biases or privacy concerns if not properly controlled. While our work is not directly tied to any specific deployment, these considerations highlight the importance of responsible use, monitoring, and safeguards when applying continual learning systems in real-world settings.

Overall, this work contributes to a better understanding of learning dynamics in non-stationary environments, which can support the development of more robust and reliable adaptive systems.

A.2 Limitations

Our work has several limitations.

First, QOA relies on a support set for prediction, which may introduce memory and computation constraints in large-scale settings,

Second, the algorithm in its current form is supposed to handle loss of plasticity and not catastrophic forgetting when task ids are unknown.

Third, while we evaluate on a range of benchmarks including Permuted MNIST, Slowly Changing Regression, CIFAR-100, Tiny ImageNet, and ImageNet, these are still controlled settings. Further validation on more complex, real-world continual learning scenarios is needed to assess generality.

Fourth, our theoretical analysis provides intuition for the emergence of a global solution through task-invariant constraints, but does not constitute a formal proof. A more rigorous theoretical characterization of this phenomenon remains an important direction for future work.

Finally, while QOA achieves strong prequential performance and low regret, its behavior in settings with extreme class imbalance or rapidly shifting label spaces remains to be explored.

B Task-Level Representations and Derivation of Global Solutions

Setup. In online continual learning, the data stream is organized as a sequence of tasks. Let τ denote the task index and i denote the datapoint index within task τ . The supervised objective can be written as

$$\min_{\theta} \sum_{\tau=1}^T \sum_{i=1}^{n_{\tau}} \mathcal{L}(f_{\theta}(x_{\tau,i}), y_{\tau,i}), \quad (2)$$

where $x_{\tau,i}$ and $y_{\tau,i}$ are the input and label of datapoint i in task τ .

Predictions are computed using a buffer

$$B = \{(x_j, y_j)\}_{j=1}^{|B|}.$$

For clarity, assume that the current datapoint has label k , denoted $x_{\tau,t}^k$, and that the support set contains one sampled support example per label from the buffer. Let x_j^{ℓ} denote the support datapoint sampled from the buffer with label ℓ .

Query-only task representation. In QOA, the task-specific representation for query $x_{\tau,t}^k$ is the vector of learned query-support scores:

$$\theta_{\tau,t}^k = [\theta_{\tau,t}^{k,1}, \theta_{\tau,t}^{k,2}, \dots, \theta_{\tau,t}^{k,C}], \quad \theta_{\tau,t}^{k,\ell} \in \mathbb{R}, \quad (3)$$

where C is the number of labels in the task. Each component $\theta_{\tau,t}^{k,\ell}$ measures the relationship between the current data-point $x_{\tau,t}^k$ and a support data-point x_j^{ℓ} with label ℓ assuming a classification task.

Since t only is only an index representing current timestep, we drop it when discussing the task-level structure and write

$$\theta_{\tau}^k = [\theta_{\tau}^{k,1}, \dots, \theta_{\tau}^{k,C}].$$

Prediction by label aggregation. For classification, QOA predicts by aggregating support labels using the entries of θ_{τ}^k . Assuming one support example per label for simplicity, the prediction can be written as

$$Y_{\text{pred}} = \sum_{\ell=1}^C \theta_{\tau}^{k,\ell} e_{\ell}, \quad (4)$$

where e_{ℓ} is the one-hot vector for label ℓ . Equivalently,

$$Y_{\text{pred}} = \theta_{\tau}^k Y^{\text{sup}}, \quad (5)$$

where $Y^{\text{sup}} \in \mathbb{R}^{C \times C}$ contains the one-hot labels of the sampled supports.

Constraint induced by training. For the current datapoint with label k , the correct prediction requires the coefficient multiplying the correct one-hot label e_k to be larger than the coefficients multiplying incorrect labels. Thus, training induces the constraint

$$\theta_{\tau}^{k,k} > \theta_{\tau}^{k,\ell}, \quad \forall \ell \neq k, \quad \ell \in \{1, \dots, C\}. \quad (6)$$

This constraint is not task-specific: it must hold for every task τ . Therefore, for all tasks, the representation induced by QOA must satisfy the same label-based ordering:

$$\theta_{\tau}^{k,k} > \theta_{\tau}^{k,\ell}, \quad \forall \ell \neq k, \quad \forall \tau. \quad (7)$$

Global solution. Because the same constraint is shared across tasks, the model is driven toward parameters θ^* that satisfy this constraint for all tasks 2a. Once this global constraint is satisfied, the learned representation no longer needs to change substantially from task to task. We refer to this regime as a *global solution*. At this point, the repeated updating of θ across tasks is reduced 2b, which mitigates the continual parameter drift that contributes to loss of plasticity.

Extension to full attention. The same argument applies to full attention with one modification. In QOA, the component $\theta_{\tau}^{k,\ell}$ is produced directly by the learned query–support scoring function. In full attention, this component is instead computed through the transformed dot product.

$$\theta_{\tau}^{k,\ell} = (b_Q(x_{\tau,t}^k))^{\top} b_K(x_j^{\ell}), \quad (8)$$

where $b_Q(\cdot)$ and $b_K(\cdot)$ are the learned query and key transformations.

The remaining derivation is unchanged: the prediction is still formed by aggregating support-label information using query–support scores, and the learning objective still induces the constraint

$$\theta_{\tau}^{k,k} > \theta_{\tau}^{k,\ell}, \quad \forall \ell \neq k, \quad \forall \tau. \quad (9)$$

Thus, full attention can also satisfy the same global constraint. However, because it uses additional query/key transformations and support interactions, the route to satisfying this constraint is less direct than in QOA.

Summary We refer to this regime as a *global solution*, where a single task-level representation explains all tasks. In this regime, predictions become stable across tasks, reducing the need for continual adaptation and mitigating loss of plasticity.

B.1 Gradient-Based Adaptation: MAML in Online Continual Learning

To isolate the role of the *global solution*, we study a model that also produces task-specific parameters but does not enforce a shared structure across tasks. This allows us to distinguish whether high performance arises from task-specific adaptation alone, or from the alignment of such adaptations.

We adapt Model-Agnostic Meta-Learning (MAML) to the online continual learning setting by using a replay buffer to simulate task structure. At each step, the buffer is partitioned into pseudo-tasks, from which support and query sets are sampled to perform inner and outer updates.

However, MAML is designed for stationary task distributions and requires multiple samples per task to compute stable gradients. In the fully online setting considered here, the data distribution is non-stationary and each sample is observed only once, making gradient-based adaptation unstable under distributional shift.

Although MAML produces task-specific parameters of the form

$$\theta'_{\tau} = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\tau}(\theta), \quad (10)$$

these parameters are entirely driven by task-specific gradients and do not impose any task-invariant structural constraint.

Consequently, the resulting representations θ'_{τ} are not required to align across tasks, preventing convergence to a shared global solution. This leads to continual parameter drift and results in loss of plasticity despite the presence of task-specific adaptation.

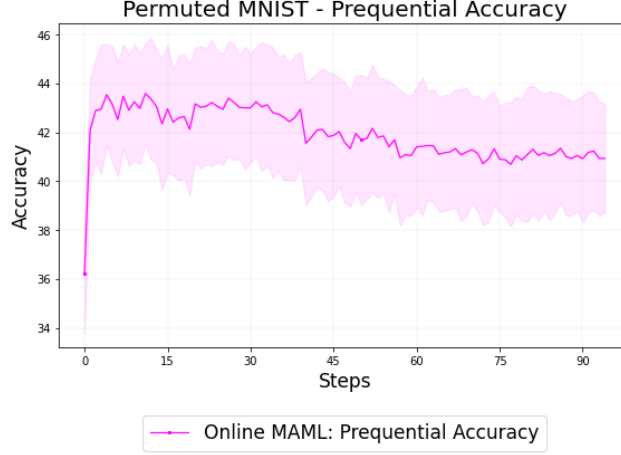


Figure 4: Prequential accuracy on Permuted MNIST. MAML exhibits degraded performance under non-stationary data, indicating loss of plasticity.

Algorithm 2 MAML with Replay Buffer

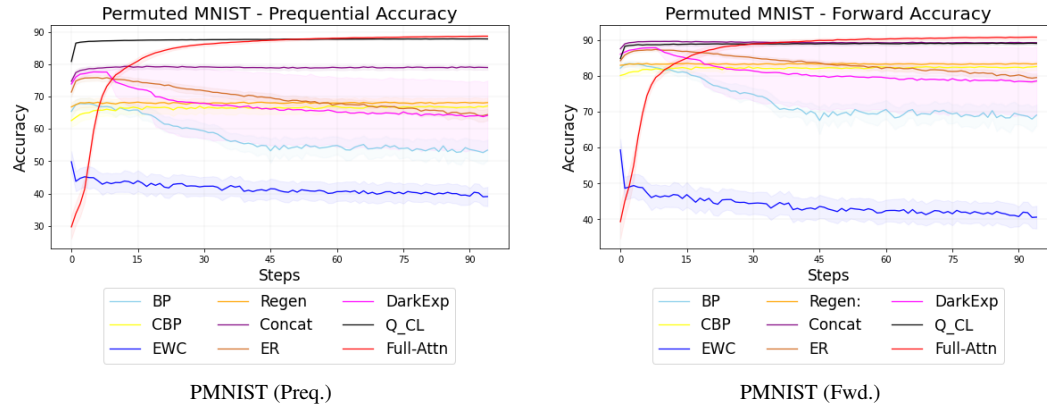
Input: Stream (x_t, y_t) ; buffer $\mathcal{B}$ of size N ; support size s ; query size q ; learning rates α, β

- 1: Initialize θ ; $\mathcal{B} \leftarrow \emptyset$
 - 2: **for** each incoming (x_t, y_t) **do**
 - 3: Insert (x_t, y_t) into $\mathcal{B}$; evict oldest if $|\mathcal{B}| > N$
 - 4: Partition $\mathcal{B}$ into pseudo-tasks
 - 5: **for** each pseudo-task **do**
 - 6: Sample support set $\mathcal{S}$ and query set $\mathcal{Q}$
 - 7: $\theta' = \theta - \alpha \nabla_{\theta} \mathcal{L}(\mathcal{S}; \theta)$
 - 8: $\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}(\mathcal{Q}; \theta')$
 - 9: **end for**
 - 10: **end for**
-

Key insight. These results show that generating task-specific parameters alone is insufficient for robust continual learning. High performance requires that such parameters share a common structure across tasks. QOA and full attention achieve this through a task-invariant matching constraint, enabling convergence to a global solution. In contrast, MAML lacks this shared structure, leading to misaligned task adaptations and loss of plasticity.

C Additional Results

Here, we present all plots corresponding to the results in Sec. 5.2, along with additional evaluations on a held-out set. The held-out metrics are referred to as forward accuracy (for classification) and forward loss (for regression).



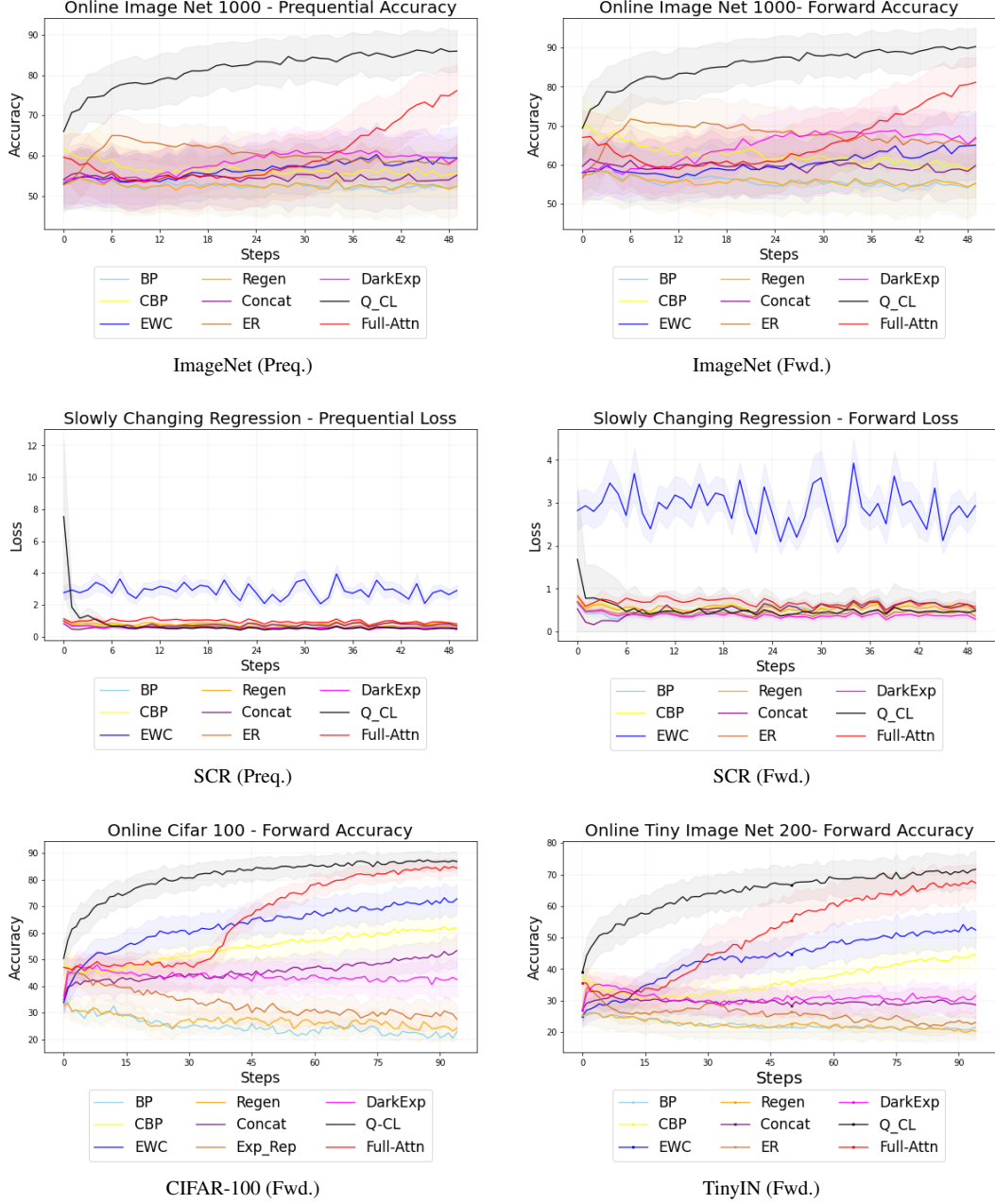


Figure. Prequential (Preq.) and forward (Fwd.) performance across datasets. Prequential curves evaluate performance before model updates, capturing online adaptation dynamics, while forward curves measure performance after adaptation within each task. Across classification benchmarks, QOA consistently achieves higher performance in both prequential evaluation and held-out accuracy, along with faster convergence. On SCR, performance differences highlight distinct optimization behavior in regression settings.

Discussion. Prequential performance reflects the model’s ability to adapt to the data stream, while forward performance captures its final accuracy after within-task updates. Across all classification datasets, QOA achieves higher prequential accuracy, indicating faster adaptation and reduced loss of plasticity. This is further supported by near-zero regret values in Table 2. In contrast, replay and regularization-based methods exhibit slower adaptation, as reflected by lower prequential performance despite competitive forward accuracy. On SCR, the behavior differs, suggesting that regression settings induce distinct optimization dynamics compared to classification tasks.

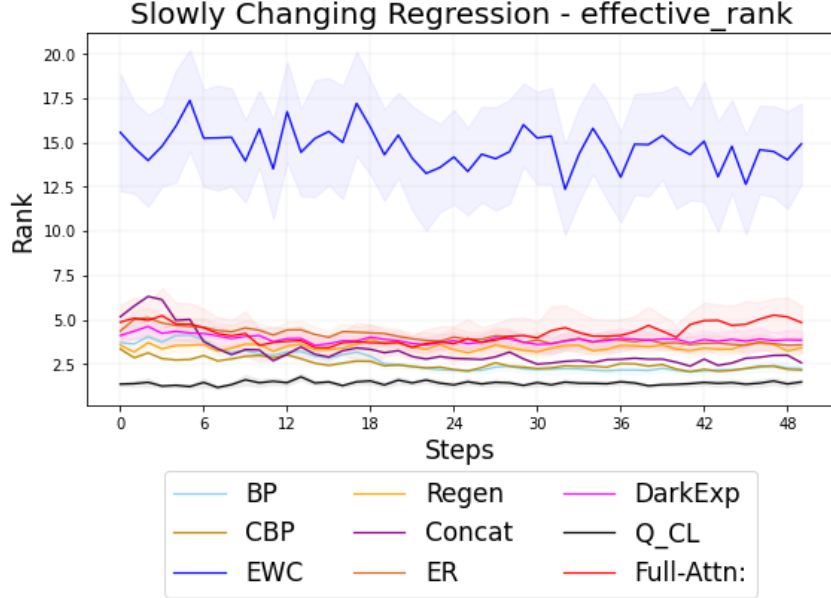


Figure 5: Effective rank dynamics across tasks. QOA maintains stable rank, indicating preserved plasticity.

C.1 Hessian rank analysis

To study plasticity, we analyze the *effective rank* of the Hessian [24, 31], defined as

$$\text{erank}(H) = \exp\left(-\sum_{i=1}^n p_i \log p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}, \quad (11)$$

where $\{\lambda_i\}$ are the eigenvalues of the Hessian. A stable non-decreasing effective rank indicates models that preserve plasticity across tasks.

The effective rank provides a measure of model plasticity. We measure it at the start of each task using the Hessian of the final layer only, since full-Hessian computation is infeasible. We evaluate it on Permuted MNIST and SCR (not ImageNet due to computational constraints). The effective rank is normalized since it depends on network size.

In both Permuted MNIST and SCR, vanilla backpropagation shows a steady drop in effective rank, aligning with loss of plasticity. However, we do not find a strong correlation between loss of plasticity and changes in Hessian rank, despite some evidence of a weak relationship.

D Implementation Details

Data generation and task-wise augmentations. For Permuted MNIST (PMNIST), each task is generated by applying a fixed random permutation to all images, creating distributional shift across tasks while preserving labels. We follow the setup of Dohare et al. [11]. For Slowly Changing Regression (SCR), we adopt the same non-stationary data generation process as Dohare et al. [11], where the underlying function evolves over time.

For CIFAR-100, Tiny ImageNet, and ImageNet, tasks are constructed by sampling a subset of classes and drawing images with replacement, enabling an effectively unbounded sequence of tasks with recurring classes. To induce controlled non-stationarity, we apply task-specific augmentations. For each task τ , augmentation parameters are deterministically generated from the task index using a hash function, ensuring reproducibility. These parameters remain fixed within a task and define the strength of each transformation, while stochasticity is introduced at the sample level (e.g., random flips, crop offsets, and noise realizations). The augmentation pipeline includes horizontal flipping, random cropping with reflection padding, additive Gaussian noise, spatial blurring via average pooling, brightness and contrast adjustments, channel-wise color shifts, desaturation toward grayscale, and cutout masking. Formally, each task corresponds to a distribution $D_\tau(x, y)$ obtained

Table 3: Task-dependent augmentation parameters. Parameters are deterministically generated per task using a hash of the task index. Transformation strengths remain fixed within a task, while stochastic components are sampled per input.

Transformation	Parameter	Range	Description
Horizontal Flip	p	0.5	Applied independently per sample
Crop (reflection pad)	p_{crop}	$\{2, \dots, 8\}$	Padding size before 32×32 crop
Gaussian Noise	σ	$[0, 0.1]$	Std. dev. of additive noise
Blur (avg pool)	b	$\{0, \dots, 4\}$	Kernel size $k = 2b + 1$
Brightness	δ_b	$[0, 0.3]$	Scaling in $[1 - \delta_b, 1 + \delta_b]$
Contrast	δ_c	$[0, 0.3]$	Scaling in $[1 - \delta_c, 1 + \delta_c]$
Color Shift	γ	$[0, 0.3]$	Channel-wise scaling in $[1 - \gamma, 1 + \gamma]$
Desaturation	α	$[0, 1]$	Interpolation with grayscale
Cutout	h	$\{0, \dots, 11\}$	Square mask size $h \times h$

by applying a transformation T_τ to the underlying dataset, where T_τ is parameterized by task-specific values for these transformations. This results in recurring classes under evolving input distributions, creating a non-stationary setting. The same transformation T_τ is applied to both training and test samples within each task. Prequential accuracy is computed on the data stream (each sample evaluated before training), and an additional held-out test set is used at the end of each task to assess robustness.

For CIFAR-100, each task samples 5 classes (5-way classification). For Tiny ImageNet, the same protocol is used over 200 classes. For ImageNet, tasks are constructed as binary (2-way) classification problems due to computational constraints.

Classes are sampled independently for each task, allowing them to reappear across tasks. At each timestep, a fixed number of datapoints is sampled uniformly from the selected classes.

Data is processed in a single pass, with each datapoint observed only once and no revisiting of past data. Although images are sampled with replacement, the task-specific augmentations make repeated samples effectively distinct, resulting in near-unique inputs across time-steps.

Hardware. All experiments were conducted on NVIDIA Ada-series GPU and a NVIDIA RTX 3090 GPU.

D.1 Experiment Specification

Table 4: Implementation details for all benchmarks. PMNIST, SCR, CIFAR-100, Tiny ImageNet, ImageNet-1000.

Parameter	PMNIST	SCR	CIFAR-100	TinyIN	ImageNet
num_tasks	10000	—	10000	10000	5000
total_classes	10	—	100	200	1000
classes_per_task	10	—	5	5	2
num_old_task_window	5	5	5	5	2
num_datapoints_per_timestep	400	1	10	10	10
num_data_points	—	10M	—	—	—
train_size	—	9000	—	—	—
flip_after	—	10000	—	—	—
change_after	60000	—	—	—	—
image_size	7 x 7	—	3 x 32 x 32	3 x 32 x 32	3 x 128 x 128

Table 5: Model hyperparameters for BP, CBP, EWC, RegenReg, ConcatRelu, ER, DER++, Full-Attn, QOA on permuted MNIST).

Parameter	BP	CBP	EWC	Regen	Concat	ER	DarkExp	Full-Attn	QOA
input_size	49	49	49	49	49	49	49	49	49
num_features	200	200	200	200	200	200	200	100	40
hidden_layers	4	4	4	4	4	4	4	–	4
attention_layers	–	–	–	–	–	–	–	3	–
step_size	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
weight_decay	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
maturity	–	100	–	–	–	–	–	–	–
decay_rate	–	0.99	–	–	–	–	–	–	–
utility_type	–	adaptable	–	–	–	–	–	–	–
replacement_rate	–	1e-4	–	–	–	–	–	–	–
ewc_lambda	–	–	100000	–	–	–	–	–	–
regenerative_lambda	–	–	–	0.01	–	–	–	–	–
buffer_size	–	–	–	–	–	2000	2000	1600	1600
buffer_type	–	–	–	–	–	reservoir	reservoir	–	–
samples_per_label	–	–	–	–	–	–	–	10	100
alpha	–	–	–	–	–	–	0.5	–	–
beta	–	–	–	–	–	–	0.5	–	–

Table 6: Model-specific hyperparameters for BP, CBP, EWC, RegenReg, ConcatRelu, ER, DER++, Full-Attn, QOA on CIFAR-100).

Parameter	BP	CBP	EWC	Regen	Concat	ER	DarkExp	Full-Attn	QOA
step_size	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
momentum	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
weight_decay	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
hidden_layer_size	100	100	–	–	100	100	100	100	50
optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
reset_head	–	false	–	–	–	–	–	–	–
early_stopping	–	true	–	–	–	–	–	–	–
replacement_rate	–	1e-5	–	–	–	–	–	–	–
utility_function	–	contribution	–	–	–	–	–	–	–
maturity_threshold	–	1000	–	–	–	–	–	–	–
noise_std	–	0	–	–	–	–	–	–	–
perturb_weights	–	false	–	–	–	–	–	–	–
ewc_lambda	–	–	1000	–	–	–	–	–	–
regenerative_lambda	–	–	–	1e-5	–	–	–	–	–
buffer_size	–	–	–	–	–	2000	2000	100	100
samples_from_buffer	–	–	–	–	–	10	10	–	–
fifo_samples_per_label	–	–	–	–	–	–	–	10	10
alpha	–	–	–	–	–	–	0.5	–	–
beta	–	–	–	–	–	–	0.5	–	–

Table 7: Model-specific hyperparameters for BP, CBP, EWC, RegenReg, ConcatRelu, ER, DER++, Full-Attn, QOA on Tiny ImageNet).

Parameter	BP	CBP	EWC	Regen	Concat	ER	DarkExp	Full-Attn	QOA
step_size	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
momentum	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
weight_decay	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
hidden_layer_size	100	100	–	–	100	100	100	100	50
optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
reset_head	–	false	–	–	–	–	–	–	–
early_stopping	–	true	–	–	–	–	–	–	–
replacement_rate	–	1e-5	–	–	–	–	–	–	–
utility_function	–	contribution	–	–	–	–	–	–	–
maturity_threshold	–	1000	–	–	–	–	–	–	–
noise_std	–	0	–	–	–	–	–	–	–
perturb_weights	–	false	–	–	–	–	–	–	–
ewc_lambda	–	–	1000	–	–	–	–	–	–
regenerative_lambda	–	–	–	1e-5	–	–	–	–	–
buffer_size	–	–	–	–	–	2000	2000	100	100
samples_from_buffer	–	–	–	–	–	10	10	–	–
fifo_samples_per_label	–	–	–	–	–	–	–	10	10
alpha	–	–	–	–	–	–	0.5	–	–
beta	–	–	–	–	–	–	0.5	–	–

Table 8: Model-specific hyperparameters for BP, CBP, EWC, RegenReg, ConcatRelu, ER, DER++, Full-Attn, QOA on ImageNet-1000).

Parameter	BP	CBP	EWC	Regen	Concat	ER	DarkExp	Full-Attn	QOA
step_size	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
momentum	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
weight_decay	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
hidden_layer_size	100	100	–	–	100	100	100	100	50
optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
reset_head	–	false	–	–	–	–	–	–	–
early_stopping	–	true	–	–	–	–	–	–	–
replacement_rate	–	1e-5	–	–	–	–	–	–	–
utility_function	–	contribution	–	–	–	–	–	–	–
maturity_threshold	–	1000	–	–	–	–	–	–	–
noise_std	–	0	–	–	–	–	–	–	–
perturb_weights	–	false	–	–	–	–	–	–	–
ewc_lambda	–	–	1000	–	–	–	–	–	–
regenerative_lambda	–	–	–	1e-5	–	–	–	–	–
buffer_size	–	–	–	–	–	2000	2000	100	100
samples_from_buffer	–	–	–	–	–	10	10	–	–
fifo_samples_per_label	–	–	–	–	–	–	–	10	10
alpha	–	–	–	–	–	–	0.5	–	–
beta	–	–	–	–	–	–	0.5	–	–

Table 9: Model-specific hyperparameters for BP, CBP, EWC, RegenReg, ConcatRelu, ER, DER++, Full-Attn, QOA on slowly changing regression).

Parameter	BP	CBP	EWC	Regen	Concat	ER	DarkExp	Full-Attn	QOA
num_inputs	20	20	20	20	20	20	20	20	41
num_outputs	1	1	1	1	1	1	1	1	1
num_features	40	40	40	40	40	40	40	14	20
activation	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU	ReLU
step_size	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
weight_decay	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
optimizer	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam	Adam
β_1	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
β_2	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
ewc_lambda	–	–	100000	–	–	–	–	–	–
regenerative_lambda	–	–	–	0.01	–	–	–	–	–
buffer_size	–	–	–	–	–	2000	2000	–	–
fifo_buffer_size	–	–	–	–	–	–	–	100	100
samples_from_buffer	–	–	–	–	–	1	1	–	–
fifo_samples	–	–	–	–	–	–	–	100	100
alpha	–	–	–	–	–	–	0.5	–	–
beta	–	–	–	–	–	–	0.5	–	–

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Answer:

Answer: [No]

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